HACKINSIGHT 2026

FloodLens Pakistan

District heavy-rain warnings 1–3 days ahead, with a weekly food-price watch

Team Spark H | Climate & disaster risk · food security

huggingface.co/spaces/RoshaanT1/floodlens-pakistan



160 districts · scored every hour

Problem Statement

THE PROBLEM

Monsoon rain in Pakistan turns deadly within hours. District planners must decide where to send rescue teams, pumps and relief before the rain arrives, but weather models give grid cells, not a per-district answer to “is this unusual for us?”

1,037

deaths and 1,067 injured in the 2025 monsoon (NDMA final sitrep No. 87)

1

The decision window

District and PDMA planners need 24–72 h to pre-position Rescue 1122 teams, pumps and relief before roads wash out

2

The gap today

Alerts are provincial (“heavy rain across KP”); no calibrated per-district chance of rain that is unusual for that district

3

The compound threat

Floods cut supply routes and spike staple prices; disaster response and market watch work in silos

Dataset Declaration

Dataset name	FloodLens district-day warehouse (5 public sources joined on 160 districts)
Source / link	Open-Meteo Previous Runs (API) · NASA GPM IMERG V07 · OCHA boundaries (HDX) · PBS weekly SPI · NDMA sitreps
Size	476,640 forecast rows × 25 features · 689,280 satellite rain rows · 146,523 prices · 157 NDMA documents
Time period	Satellite rain 2015–2026 · forecasts Feb 2024–Oct 2026 · prices Jan 2023–Oct 2026
Format & license	4,856 raw files in manifest.csv (source, query, UTC time, SHA-256, licence) → Parquet / DuckDB · CC BY 4.0, NASA open data, public govt. statistics
Key features / target	Forecast rain, humidity, pressure, cloud, wind (1–3 days ahead), lagged observed rain. Target: rain above the district's own 95th percentile

476K
Rows / records (model table)

25
Columns / features

1.3 GB
Raw data, 4,856 files

Why These Sources: Forecasts vs Satellite Truth

INPUTS (X) · OPEN-METEO ARCHIVED FORECASTS

- Previous Runs API: ECMWF, NOAA and DWD forecasts exactly as issued 1, 2 and 3 days ahead
- Rain sum, max hourly rain, humidity, pressure, cloud cover, wind gusts
- Honest lead time: the model only sees forecasts that existed before the target day

TRUTH (Y) · NASA GPM IMERG V07 SATELLITE RAIN

- 0.1° daily satellite rainfall, 2015–2026: labels and each district's 95th-percentile threshold (2015–2023)
- Why not gauges? Pakistan's gauges are sparse in the remote, flood-prone districts; satellites cover all 160
- Independent of the weather models, so the model is not graded against its own inputs

Pipeline Used

1

Data Collection

APIs, NASA cloud, PBS and NDMA files; every file hashed into a manifest

2

Data Cleaning

Quality agent: fix, fill or flag each issue, with the reason logged

3

Analysis / EDA

District thresholds, monsoon replay, error by province and terrain

4

Modeling

Gradient boosting, calibrated; beats climatology on the locked test set

5

Dashboard / App

Hourly Prefect flow publishes to a live Hugging Face Space

Raw forecasts and satellite rain in → each district's chance of unusually heavy rain 1–3 days ahead, refreshed every hour

Pipeline Details & Tools

Stage	Tools / Libraries	What we did
Collection	Python, requests, earthaccess (OPeNDAP), pdfplumber	Fetch only what is new; respect API budgets; never overwrite a raw file
Preprocessing	DuckDB, pandas, Shapely, pandera	Grid rain → district values; quality rules fix / fill / flag and log every decision
Analysis / EDA	pandas, bootstrap CIs	95th-percentile thresholds per district; Aug 2025 replay; skill by province and terrain
Modeling	scikit-learn GBM + Platt calibration, FAISS + e5	Leakage-tested features; fixed splits; test set run once. Insight Agent cites NDMA text
Dashboard / App	Prefect 3, React + Leaflet, FastAPI, HF Spaces	Hourly flow rebuilds and publishes; map, food prices, live data and Ask (English / Urdu)

Agents & Automation

DATA QUALITY AGENT · runs pandera checks, decides fix / fill / flag, logs why

Rule	Action	Rows	Reason (from quality_log)
spi_price_collected	FLAG	1,279	PBS published 0 or nothing: kept NULL, never filled
spi_min_le_avg_le_max	FLAG	822	min / avg / max out of order as published: kept as printed
spi_weekly_jump	FLAG	399	moved >50% in a week: possible typo, not corrected

Every decision is stored with a timestamp and replayed on each rebuild; we flag rather than invent prices.

Insight Agent · retrieves NDMA sections (FAISS, multilingual) and gold-table rows; says "I don't know" below threshold.

HOURLY PREFECT FLOW

cron 5 * * * *

15 runs in the Tue 4 PM – Wed 4 PM window

14 completed

1 failed: NASA server refused us; IMERG step now isolated, switched to NASA cloud

Optimization: Lean, Fast & Reproducible

143 GB → 182 MB

SERVER-SIDE CROPPING

NASA OPeNDAP crops each daily satellite file to Pakistan: 33 MB → ~74 KB per day, same values

286 s → 3.6 s

PARSE CACHE

PBS price PDFs parsed once and cached by SHA-256; identical tables on every rebuild

Fail-safe runs

ATOMIC SWAPS

Each hourly run builds a scratch DuckDB and swaps it in only on success; a failed run never touches live data

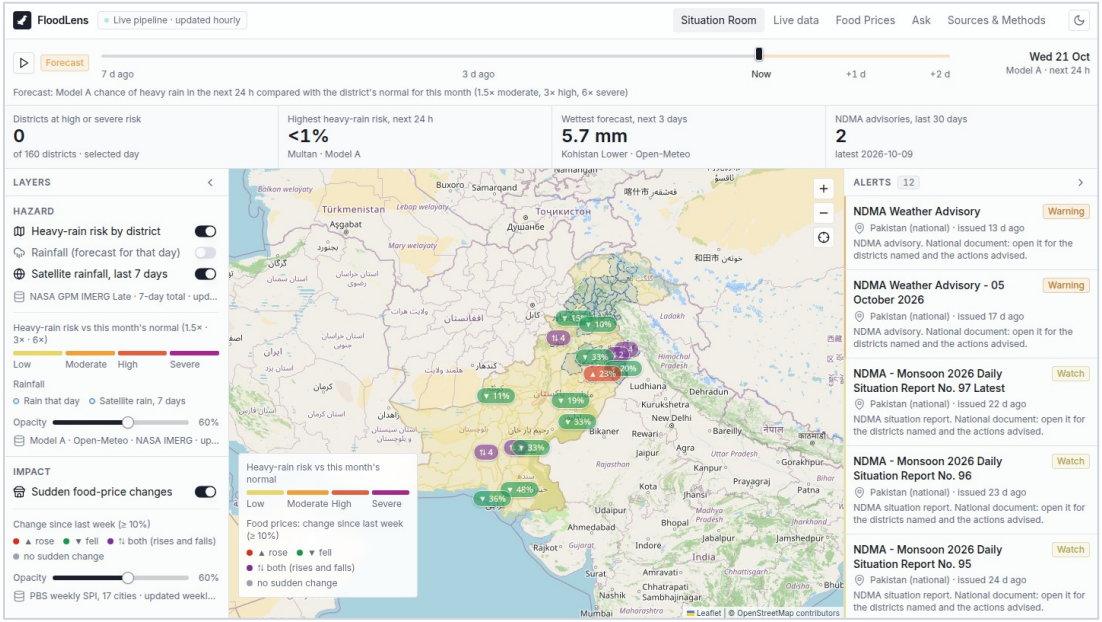
1 command

MAKE DATA

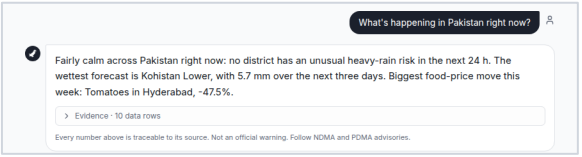
Rebuilds raw → cleaned → gold from the manifest and refuses any file whose hash changed

Dashboard / App Views

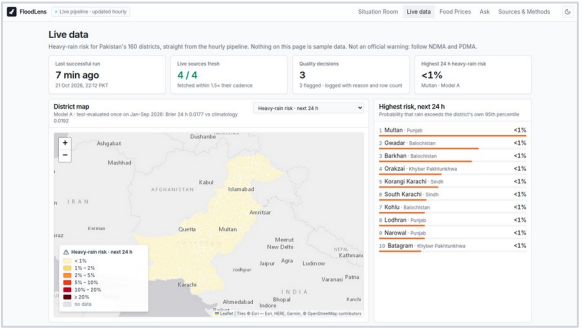
Page 1 – Situation Room: risk vs each district's normal



Page 2 – Ask (Insight Agent)



Page 3 – Live data



Live demo: huggingface.co/spaces/RoshaanT1/floodlens-pakistan • **Repo:** github.com/RoshaanT1/floodlens-pakistan

Model A: How It Works

ALGORITHM

Gradient boosting

scikit-learn HistGradientBoosting: 300 iterations, 15 leaf nodes, ≥100 samples per leaf, L2 = 1. Chosen over logistic regression on validation (Brier 0.0233 vs 0.0238).

CALIBRATION

Probabilities you can trust

Platt scaling fitted on the validation period (cross-fitted by week), so a 20% warning means rain like this about 1 time in 5.

NO LEAKAGE

Only what was known

Forecasts issued 1–3 days before the target day; satellite rain only from days that ended ≥14 h before issue. Automated tests enforce both.

Fixed time splits: train to 30 Jun 2025 · validate Jul–Dec 2025 · test Jan–Sep 2026, run once with the frozen model. Baseline declared in advance: district-by-month climatology.

Results vs Baseline

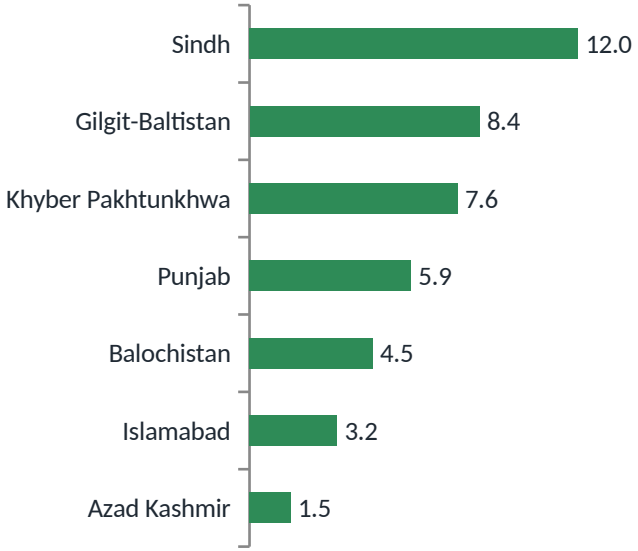
TEST SET Jan–Sep 2026, run once · Brier [95% CI], lower is better

	24 h	48 h	72 h
Model (GBM)	**0.0177 [0.0146, 0.0213]**	**0.0178 [0.0147, 0.0214]**	**0.0179 [0.0148, 0.0215]**
Climatology (declared)	0.0192 [0.0156, 0.0231]	0.0192	0.0192
Paired difference	−0.0014 [−0.0022, −0.0008]	−0.0013 [−0.0021, −0.0007]	−0.0013 [−0.0019, −0.0007]

Monsoon months: 0.0237 vs 0.0251 · Validation (model choice): GBM 0.0233, logistic 0.0238, climatology 0.0249

Time-based splits (train ≤ Jun 2025, validate Jul–Dec 2025), leakage tests on every feature, test locked in code

Better than climatology in every province (validation, 24 h, %)



Weakest: Azad Kashmir (+1.5%). Model B (prices) did not beat “no change” (MAE 1.667 vs 1.665): rejected and shown as baseline.

Responsible AI & Pakistan Impact

No invented numbers

Every number is inserted by code from a gold table and carries its row id

Every claim cited

NDMA quotes carry document, page and section id; no support → “I don't know”

Safe by default

Emergency questions go to Rescue 1122 and NDMA/PDMA first; never an official warning

Usable in Pakistan

Urdu answers, 160 OCHA districts, PBS prices for 17 cities, NDMA advisories on the map

LOCAL VALIDATION (NDMA)

- Replay of Aug 2025: on 14 Aug, 8 of our top-10 districts had heavy rain (34 districts did)
- NDMA's 87 sitreps name 125 districts; our fixed 0.5 alert rule flagged only 1, so we rank risk instead

HONEST LIMITS

Buner, 15 Aug 2025: a flash flood killed 100+, but the satellite's wettest pixel (43 mm) stayed under Buner's 57.5 mm threshold, so the label says “no heavy rain”. Daily satellite rain misses cloudbursts.

Why FloodLens: Compared with Today's Alerts

	Provincial alerts & weather apps	FloodLens Pakistan
Timing	Often reported once flooding has started	**24–72 h ahead, refreshed hourly**
Detail	Province or division (“rain in Sindh”)	**Each of 160 districts, against its own normal**
Rigour	Raw forecasts, no stated skill	**Calibrated; beats climatology on a locked test set**
Trust	Claims without sources	**Row ids for numbers, NDMA section ids for text**
Food impact	Not linked to markets	**Weekly PBS prices for 17 cities on the same map**

Business Model & Sustainability

WHO USES IT

- District disaster-management officers
- PDMAs (Punjab, KP, Sindh, Balochistan, GB, AJK) and NDMA
- NGOs that pre-position relief

WHO PAYS

- PDMA / NDMA preparedness budgets
- Donors funding anticipatory action: money released when a forecast trigger is hit
- Public dashboard stays free

WHAT IT COSTS

- One CPU: a small VM (~€4/month) or a free tier
- Open data: NASA, OCHA, PBS; Open-Meteo free for non-commercial use
- ~1.3 GB storage, +10 MB a day

Value: turns reactive relief spending into planned pre-positioning, with evidence that settles where resources go first

Conclusion

1

Beats the baseline on unseen 2026 data

Brier 0.0177 vs climatology 0.0192 (about 7% better); the 95% interval of the gain excludes zero

2

Better everywhere we checked

Ahead of climatology in every province and terrain group and in monsoon months; weakest in Azad Kashmir

3

Right when it mattered

Replaying Aug 2025: on 14 Aug, 8 of the 10 districts we ranked highest really had heavy rain

IMPACT

160 district offices get a free, hourly, per-district signal with reasons and NDMA sources.

Payer: PDMA preparedness budgets or donors' anticipatory-action funds. Runs on 1 CPU, free-tier data.

FUTURE WORK

- Sub-daily rain to catch cloudbursts like Buner
- Flood and damage targets, not only rain
- A price model that beats “no change”

Team & Credits

RT

Roshaan Tahir · CSDT-13

Team Lead · Lead Architect

- Open-Meteo forecast pipeline
- Model A training and calibration
- Insight Agent and live dashboard

HK

M Hammad Khan · CSDT-33

Lead Backend

- NASA IMERG satellite rain fetcher
- District rain, 95th-pct thresholds, labels
- Data quality checks + 26 tests

MH

M Huzaifa · CSDT-21

Lead Data Sources

- Hugging Face Space (Docker server)
- Build-and-assemble deploy script
- Deploy checklist and AI-use records

Contact: github.com/RoshaanT1/floodlens-pakistan · Data: Open-Meteo, NASA, OCHA, PBS, NDMA

Thank you!